§2.1

$x_n \to a$ as $n \to \infty$ $\xrightarrow{\text{def}}$ $f: \mathbb{N} \to \mathbb{R}$, $f(n) := a_n$

$\lim_{n\to\infty} x_n = a \in \mathbb{R} \iff \forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $|x_n - a| < \varepsilon$ as $n \geq N$.

$\{x_n\}$ conv. to a

$\Updownarrow$ ← thm

① $|x_n - a| \to 0$ as $n \to \infty$ and ② $|x_n| \to 0 \iff x_n \to 0$ as $n \to \infty$.

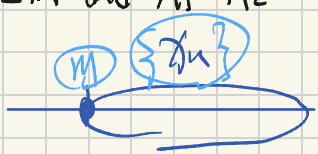
2.5 Definition.

By a *subsequence* of a sequence $\{x_n\}_{n\in\mathbf{N}}$, we shall mean a sequence of the form $\{x_{n_k}\}_{k\in\mathbf{N}}$, where each $n_k \in \mathbf{N}$ and $n_1 < n_2 < \cdots$.

$\{(-1)^n\} = \{1^k\}$

$n = 2k$

$\{x_{n_k}\}$ is subsequence of $\{x_n\} \iff \exists \{n_k\} \subseteq \mathbb{N}$ and $n_1 < n_2 < \cdots$



2.6 Remark. *If $\{x_n\}_{n\in\mathbf{N}}$ converges to a and $\{x_{n_k}\}_{k\in\mathbf{N}}$ is any subsequence of $\{x_n\}_{n\in\mathbf{N}}$, then x_{n_k} converges to a as $k \to \infty$.*

(1) $\{x_n\}$ is bounded above $\iff \exists M \in \mathbb{R}$ s.t. $x_n \leq M, \forall n \in \mathbb{N}$. (M: upper bound)

(2) $\{x_n\}$ is bounded below $\iff \exists m \in \mathbb{R}$ s.t. $x_n \geq m, \forall n \in \mathbb{N}$ (m: lower bound)

(3) $\{x_n\}$ is bounded $\iff \{x_n\}$ is bounded above and below.

2.8 Theorem. *Every convergent sequence is bounded.* (逆句不成立)

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $|x_n - a| < \varepsilon$ as $n \geq N$

$\Rightarrow -\varepsilon < x_n - a < \varepsilon$

$\Rightarrow a - \varepsilon < x_n < a + \varepsilon \Rightarrow x_n$ is bounded

($a - \varepsilon \in \mathbb{R}$, $a + \varepsilon \in \mathbb{R}$)

2.1.0. Decide which of the following statements are true and which are false. Prove the true ones and provide a counterexample for the false ones.

a) If x_n converges, then x_n/n also converges.
b) If x_n does not converge, then x_n/n does not converge.
c) If x_n converges and y_n is bounded, then $x_n y_n$ converges.
d) If x_n converges to zero and $y_n > 0$ for all $n \in \mathbf{N}$, then $x_n y_n$ converges.

$\frac{1}{n} \to 0$ as $n \to \infty$

Archimedean Principle

$a, b \in \mathbb{R}$, $a > b$

$\exists N \in \mathbb{N}$ s.t. $b < na$.

(a) If x_n converges, $\exists M > 0$ s.t. $|x_n| \le M$

By Archimedean Principle, take $N \in \mathbb{N}$ s.t.

$M < \varepsilon N \Rightarrow N > \frac{M}{\varepsilon}$

Then if $n \ge N$, $\left|\frac{x_n}{n}\right| \le \left|\frac{M}{n}\right| \le \left|\frac{M}{N}\right|$

$< M \cdot \frac{\varepsilon}{M} = \varepsilon$

$\Rightarrow \left|\frac{x_n}{n}\right| < \varepsilon \Rightarrow \frac{x_n}{n}$ conv.

$\Rightarrow$ true. //

$\left|\frac{x_n}{n}\right| < \varepsilon$ as $n \ge N$

$\left|\frac{x_n}{n}\right| \le \left|\frac{M}{n}\right| < \varepsilon$.

$\Rightarrow \frac{M}{n} < \varepsilon$.

$\Rightarrow \frac{M}{n} \le \frac{M}{N} < \varepsilon$.

$\Rightarrow M < N\varepsilon$.

(c) Let $x_n = 1^n$ and $y_n = (-1)^n$, thus $x_n y_n = (-1)^n$

(x_n converges; y_n bounded)

Since $x_n y_n = (-1)^n$ does not converge, $\lim_{n\to\infty} x_n y_n$ does not exist.

Hence it is false. //

2.1.1. Using the method of Example 2.2i, prove that the following limits exist.

(d) $\frac{3n^2-1}{6n^2} \to \frac{1}{2}$ as $n \to \infty$

pf Let $\varepsilon > 0$, $\exists N \in \mathbb{N}$ and take $N > \frac{1}{\sqrt{6\varepsilon}}$.

If $n \geq N$,

$$\left|\frac{3n^2-1}{6n^2} - \frac{1}{2}\right| = \left|\frac{-1}{6n^2}\right| \leq \frac{1}{6N^2} < \frac{1}{6} \cdot 6\varepsilon = \varepsilon$$

Hence $\frac{3n^2-1}{6n^2} \to \frac{1}{2}$ as $n \to \infty$. //

$\frac{1}{n} \to 0$ as $n \to \infty$

take $N > 0 \Rightarrow \varepsilon < 1 \cdot N$

as $n \geq N$ $\frac{1}{n} \leq \frac{1}{N} < \frac{1}{\varepsilon}$

$$\left|\frac{3n^2-1}{6n^2} - \frac{1}{2}\right| = \left|\frac{3n^2-1-3n^2}{6n^2}\right| = \left|\frac{-1}{6n^2}\right| = \frac{1}{6}\left|\frac{1}{n^2}\right| \leq \frac{1}{6N^2} < \varepsilon$$

$\Rightarrow N^2 > \frac{1}{6\varepsilon}$

$\Rightarrow N > \frac{1}{\sqrt{6\varepsilon}}$

2.1.2. Suppose that x_n is a sequence of real numbers that converges to 1 as $n \to \infty$. Using Definition 2.1, prove that each of the following limits exists.

$x_n \to 1$ as $n \to \infty$

(b) $\frac{3x_n - \pi}{x_n} \to 3 - \pi$ as $n \to \infty$

Pf ① Given $\varepsilon > 0$. Since $x_n \to 1$ as $n \to \infty$,
for $\varepsilon_1 = \frac{1}{2}$, $\exists N_1 \in \mathbb{N}$ s.t. $|x_n - 1| < \frac{1}{2}$
as $n \geq N_1$

$\Rightarrow \frac{1}{2} < x_n < \frac{3}{2} \Rightarrow |x_n| > \frac{1}{2} \Rightarrow 2 > \frac{1}{|x_n|}$

② For $\varepsilon_2 = \frac{\varepsilon}{2\pi} > 0$, since $\lim_{n\to\infty} x_n = 1$,
$\exists N_2 \in \mathbb{N}$ s.t. $|x_n - 1| < \frac{\varepsilon}{2\pi}$ as $n \geq N_2$

③ Take $N = \max\{N_1, N_2\}$, thus

$$\left|\frac{3x_n - \pi}{x_n} - (3 - \pi)\right| = \pi \frac{|x_n - 1|}{|x_n|} < \pi \cdot \frac{\varepsilon}{2\pi} \cdot 2 = \varepsilon$$

Hence $\frac{3x_n - \pi}{x_n} \to 3 - \pi$ as $n \to \infty$.

Let $\varepsilon = \frac{1}{2}$.

$|x_n - 1| < \frac{1}{2}$

$\Rightarrow \frac{1}{2} < x_n < \frac{3}{2}$

$\Rightarrow |x_n| > \frac{1}{2}$

① $\Rightarrow 2 > \frac{1}{|x_n|}$

$$\left|\frac{3x_n - \pi}{x_n} - (3 - \pi)\right| = \left|\frac{3x_n - \pi - 3x_n + \pi x_n}{x_n}\right| = \left|\frac{\pi x_n - \pi}{x_n}\right| = \pi \left|\frac{x_n - 1}{x_n}\right|$$

$\Rightarrow \pi \frac{|x_n - 1|}{|x_n|} < 2\pi |x_n - 1| < \varepsilon$

② $\Rightarrow |x_n - 1| < \frac{\varepsilon}{2\pi}$

2.1.3. For each of the following sequences, find two convergent subsequences that have different limits.

a) $2(-1)^n$

(a) Let $n_1 = 2n$, $n_2 = 2n+1$,

then $x_{n_1} = 2(-1)^{2n} = 2 \cdot 1^n$

and $x_{n_2} = 2(-1)^{2n+1} = -2 \cdot 1^n$

Thus $x_{n_1} \to 2$ and $x_{n_2} \to -2$ as $n \to \infty$.

2.1.4. Suppose that $x_n \in \mathbf{R}$.

a) Prove that $\{x_n\}$ is bounded if and only if there is a $C > 0$ such that $|x_n| \leq C$ for all $n \in \mathbf{N}$.

b) Suppose that $\{x_n\}$ is bounded. Prove that $x_n/n^k \to 0$, as $n \to \infty$, for all $k \in \mathbf{N}$.

(b) By the Archimedean Principle, take $N \in \mathbb{N}$ s.t. $N > \left(\frac{C}{\varepsilon}\right)^{1/k}$

If $n \geq N$, since $n \geq N > \left(\frac{C}{\varepsilon}\right)^{1/k}$,

then $n > \left(\frac{C}{\varepsilon}\right)^{1/k} \Rightarrow n^k > \frac{C}{\varepsilon}$

Therefore, $\left|\frac{x_n}{n^k} - 0\right| = \frac{|x_n|}{n^k} \leq \frac{C}{n^k} \leq C \cdot \frac{\varepsilon}{C} = \varepsilon$

Hence $\frac{x_n}{n^k} \to 0$ as $n \to \infty$, $\forall k \in \mathbb{N}$. //

$\frac{x_n}{n^k} \to 0$ as $n \to \infty$

$\Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $\left|\frac{x_n}{n^k}\right| < \varepsilon$ as $n \geq N$

Let $|x_n| \leq C, \forall n \in \mathbb{N}$

$\Rightarrow \left|\frac{x_n}{n^k}\right| \leq \left|\frac{C}{N^k}\right| < \varepsilon$

$\Rightarrow \frac{C}{N^k} < \varepsilon$

$\Rightarrow N^k > \frac{C}{\varepsilon}$

$\Rightarrow N > \left(\frac{C}{\varepsilon}\right)^{\frac{1}{k}}$

2.1.6. Let a be a fixed real number and define $x_n := a$ for $n \in \mathbf{N}$. Prove that the "constant" sequence x_n converges.

$\lim_{n\to\infty} x_n = a$.

pf If $x_n = a$ for all $n \in \mathbb{N}$, then $|x_n - a| = |a - a| = 0$

Therefore, $|x_n - a| = 0 < \varepsilon$, $\forall \varepsilon > 0$. and $\forall n \in \mathbb{N}$.

Hence $x_n \to a$ as $n \to \infty$.